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Insect wing Evolution

On the fly

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Insects are the first among all the creatures on the planet to take flight, and the only one to evolve specialised appendages to do so. In both the birds and bats, wings are forelimbs that have evolved for flight. One of the longstanding mysteries of science is how and when exactly did insects evolve wings. This is because the fossil record does not represent the emergence of wings, with no finds having so called proto-wings, intermediary structures between wings and whatever or whatever they evolved from. There are plenty of insect fossils from the wet, warm, swampy regions from the Carboniferous, 360 million years ago including the order Palaeodictyoptera, that had two pairs of large wings, similar to those of dragonflies. In the older insect fossils, the wings are missing, but the fossils are so few that scientists cannot pinpoint when insect wings evolved. For example, the oldest insect fossil known is from around 400 million years ago, and is only a part of a head. Still, there are well-preserved mouth parts, but these are specialised mandibles associated with winged insects, so it may well be that the winged insects had emerged in the Devonian itself. Scientists simply do not know.

Over the years, there have been many theories suggested. One was



that the insects grew fins for catching heat, as the insect wings are great at trapping the energy. Evolution nudged them towards growing larger and larger, till they were big enough for flight, but at that point the started dissipating too much heat. According to yet another theory, wings evolved originally as a form of sexual competition, with larger wings attracting more mates till they were big enough to fly. In the 90s scientists honed in on gills as a likely origin point for insect wings for a number of reasons. The gills have muscles, neural structures and joints just like the wings, and in the larval stage of some insects the gills even look like the wings. All these theories for insect wing evolution could be sorted into two buckets. The first one, that insect wings was another random mutation in an ancient ancestor that stuck, a freak of evolution, wings just started to grow out of the dorsal body walls from

little blobs of tissue early in the insect life cycles, is called the tergal origin hypothesis. The theory that the wings evolved from gills is one of the many pleural origin hypotheses, according to all of which the wings of the insect evolved from the lateral body wall tissues, and their branches, which are known as excites. Basically, the wings grew out of the back or out of the side, and it was up to the scientists to figure out which.

TERGAL AND PLEURAL ORIGIN HYPOTHESIS

With the advancement of molecular biology and genetics, new techniques began to be available to test these various theories. In 1997, researchers reasoned that if the wings of insects did in fact originate from gills, then the gills and the wings would both have some genes in common. To test this hypothesis, researchers used the polymerase chain reaction (PCR)